Vectorized Interdisciplinary Overlap between STEM Domains

ASEE 2026 Annual Conference

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Multidisciplinary Motivation

- Students exercise *agency* in course selection beyond their declared major
- Cross-departmental enrollment signals overlap between programs
- NSF identifies convergence research as a top 10 BIG Idea: grand challenges "will not be solved by one discipline alone"
- Interdisciplinary programs can broaden participation for students with marginalized identities

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Frame the problem: students are already interdisciplinary in their behavior—we want to measure and characterize that behavior structurally.

What is MIDFIELD?

- Multiple-Institution Database for Investigating Engineering Longitudinal Development
- Academic records of 98,000+ students from 18 US universities
- Data from 2000 onward to reflect contemporary curricula
- Contains: degree records, course enrollment, grades, demographics

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Emphasize that MIDFIELD has been used primarily for student persistence and migration studies. We are using it differently—to characterize relationships between programs rather than student pathways.

Contribution to the Field

- Shift unit of analysis from *student pathways* to *program relationships*
- Student enrollment as expression of agency and interdisciplinary interest
- Complementary similarity metrics applied to enrollment vectors
- Define three interpretable relationship types:
  - Nested: one program's coursework is largely contained within another
  - Parallel: shared base, divergent specialization
  - Convergent: deep overlap in both general and specialized content
- Extend curricular analytics *across* disciplinary boundaries

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Prior MIDFIELD work focuses on where students go. We focus on how programs relate to each other, using student behavior as the evidence.

Research Question

RQ: What types of curricular relationships have emerged between undergraduate STEM programs as evidenced by student enrollment patterns?

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Single research question. The methodology section explains how we operationalize "types of curricular relationships."

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From Student Records to Program Vectors

1. Identify all courses taken by students completing each CIP code
2. Count course frequencies across all graduates of that major
3. Result: a high-dimensional *enrollment vector* per program
4. Apply similarity metrics to compare vectors pairwise

From Student Records to Program Vectors

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Key insight: we go beyond binary course presence (which only enables Jaccard) to frequency data, which enables the full suite of metrics.

Why Multiple Metrics?

- A single metric cannot distinguish:
  - Broad gen-ed overlap vs. specialized content overlap
  - Subset containment vs. symmetric overlap
  - Expected vs. statistically surprising co-enrollment

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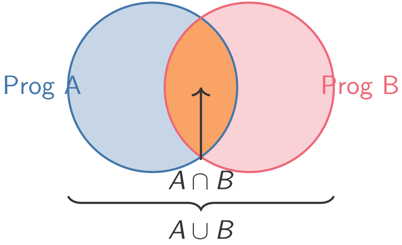
This is the methodological argument of the paper—no single number tells the whole story. Each metric answers a different question about the relationship.

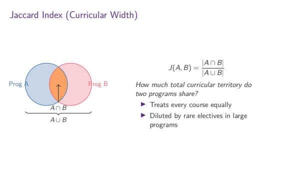
Jaccard Index (Curricular Width)



$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

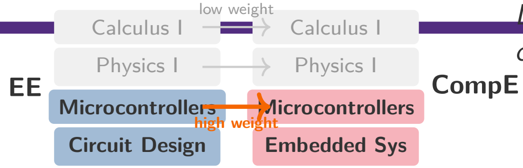
How much total curricular territory do two programs share?

- Treats every course equally
- Diluted by rare electives in large programs



Jaccard is our baseline—simple set overlap. The Venn diagram shows that the denominator is the full union, so programs with large elective sets get lower scores even if their cores overlap.

TF-IDF Cosine (Specialized Identity)



Upweights courses unique to few programs; downweights ubiquitous ones

Do two programs share identity-defining coursework?

- Resistant to program size differences
- Isolates specialized content from gen-ed noise

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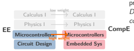

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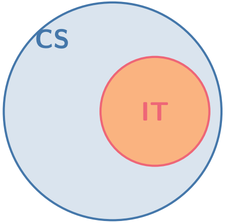
TF-IDF is the key complement to Jaccard. The diagram shows that shared gen-ed courses (gray, low weight) are ignored while shared specialized courses (colored, high weight) drive the score.

Overlap Coefficient (Subset Detection)

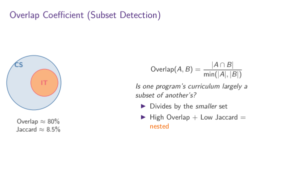


Overlap $\approx$ 80%
Jaccard $\approx$ 8.5%

$$\text{Overlap}(A, B) = \frac{|A \cap B|}{\min(|A|, |B|)}$$

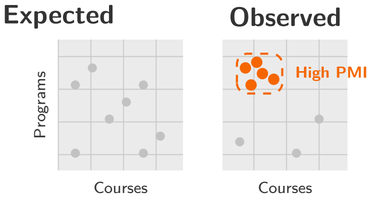
Is one program's curriculum largely a subset of another's?

- Divides by the *smaller* set
- High Overlap + Low Jaccard = nested



The diagram shows IT almost entirely inside CS. Overlap catches this because it divides by the smaller set (IT). Jaccard misses it because the union (all of CS) is so much larger.

PMI (Signature Bonds)



$$\mathrm{PMI}(m, c) = \log \frac{p(m, c)}{p(m)\, p(c)}$$

Do two programs share statistically surprising courses?

- Compares observed co-occurrence to expected
- High PMI = courses that “belong together”

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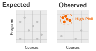

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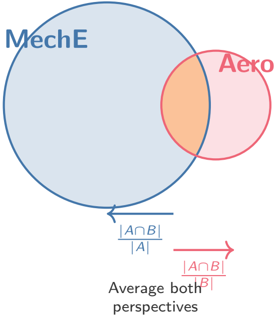
PMI detects courses that co-occur in two programs far more than chance predicts. The left panel shows uniform distribution (what we’d expect if independent). The right shows clustering—courses that concentrate in specific program pairs have high PMI.

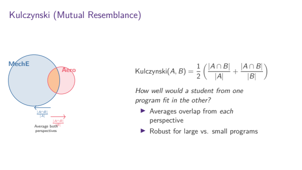
Kulczynski (Mutual Resemblance)



$$\text{Kulczynski}(A, B) = \frac{1}{2}\left(\frac{|A \cap B|}{|A|} + \frac{|A \cap B|}{|B|}\right)$$

How well would a student from one program fit in the other?

- Averages overlap from *each* perspective
- Robust for large vs. small programs



Kulczynski is important because MIDFIELD has huge enrollment imbalances. The diagram shows that it averages two different views: what fraction of MechE is shared, and what fraction of Aero is shared. This is fairer than using just one denominator.

Relationship Taxonomy

Table: Metric signatures for three curricular relationship types

Type	Jaccard	TF-IDF	Overlap	PMI	Example
Nested	Low	Low	High	Var.	IT $\subset$ CS
Parallel	Med	Low	Med	Low	EE $\parallel$ Civil Eng
Convergent	High	High	High	High	Aero $\approx$ Ind Eng

- The *pattern* across metrics defines the relationship, not any single score
- Relative values within a domain matter more than absolute percentages

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This table is the core framework. Walk through each row: nested = high overlap but low Jaccard means containment. Parallel = shared base, divergent specialization. Convergent = high everything means deep structural alignment.

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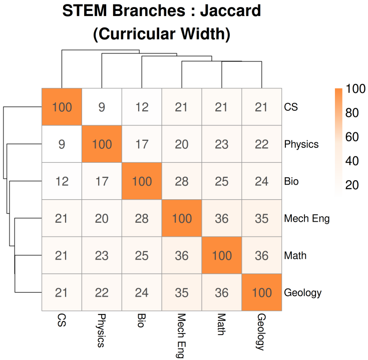
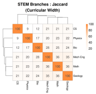
STEM Overview: Mathematics as a Bridge



- Math shares coursework with Mech Eng (~36%), Physics (~23%), CS (~21%)
- *Generalist connector* across STEM
- Validates the approach

STEM Overview: Mathematics as a Bridge



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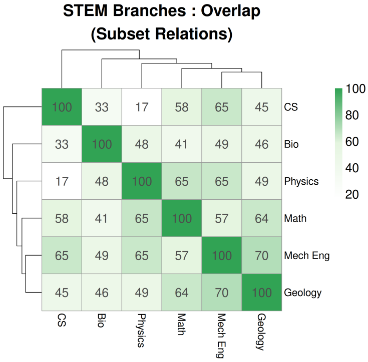
This slide grounds the audience. These relationships are intuitive—Math is indeed central to STEM. The method works.

STEM Overview: Nested Patterns Emerge



- Physics–Mech Eng: ~65% Overlap vs. ~20% Jaccard
- High Overlap + Low Jaccard = nested relationship
- Physics coursework is largely contained within Mech Eng
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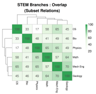

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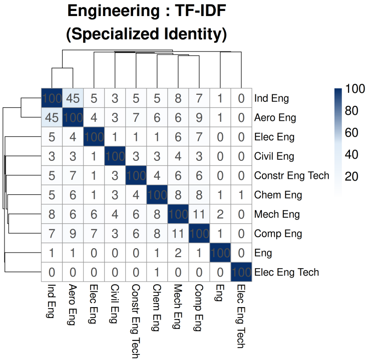
Compare the two heatmaps. Jaccard says Physics and MechEng are somewhat related. Overlap reveals the structural truth: Physics is nested within MechEng's broader curriculum.

Engineering: The Parallel Pattern



- Median TF-IDF across engineering: ~3%
- Engineering programs share a large common core…
- …but specialized content is *very* different
- This is the parallel pattern: shared base, divergent identity
- Exception: EE–CompE share ~33% Jaccard + high TF-IDF (merged departments)

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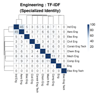

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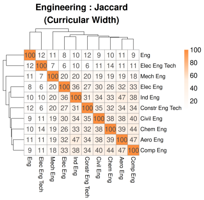
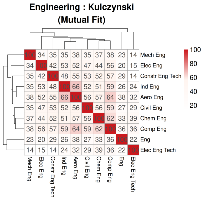
The general engineering model dominates. Shared math, physics, intro engineering—but once you get to upper-division courses, each program is distinct. The low TF-IDF median is the baseline that makes the next result so striking.

Engineering: Aerospace–Industrial Convergence





- Flagship result: Aero–Industrial scores highest on *every* metric:
  - TF-IDF ~45%, Jaccard ~47%
  - Kulczynski ~66%, Overlap ~79%
- Textbook convergent relationship
- No one groups these administratively

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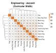

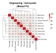

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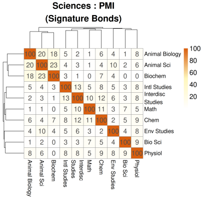
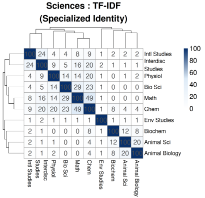
This is the result the audience should remember. Two programs that no one would group together are the most similar pair in all of engineering on every metric. Shared systems analysis and applied statistics content drives the similarity.

Sciences: Cross-CIP Clustering





- Animal Sci (CIP 01) clusters with Biochemistry & Animal Bio (CIP 26)
- Strongest PMI: Animal Sci–Biochem (∼23%)
- Crosses CIP family boundaries
- Math–Chem TF-IDF (∼49%): strongest specialized pair

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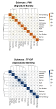

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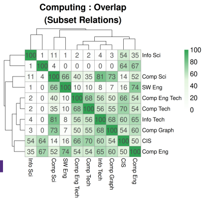
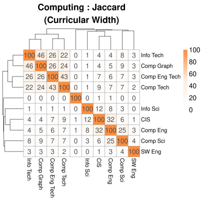
PMI is the star metric here. It finds that these three programs from different CIP families share courses at rates far above chance. The big program (Bio Sci) actually has the weakest surprising connections—its breadth dilutes specificity.

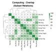
Computing: The Nested Hierarchy

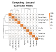




- IT–CS: cleanest nested example in the dataset
  - Overlap: ~80%
  - Jaccard: ~8.5%
  - Largest divergence between these metrics anywhere
- CS–CompE: mixed nested/convergent (Overlap ~52%, TF-IDF ~29%)
- Computing is organized as *nested hierarchies* branching from CS

Computing: The Nested Hierarchy





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If you only looked at Jaccard, you'd say IT and CS are barely related. Overlap reveals the truth: nearly all IT coursework is contained within CS. This is why single-metric analysis is insufficient.

Interactive Explorer

- All metrics, domains, and program pairs
- Heatmaps with dendrograms
- Computing, engineering, sciences, cross-domain

https://ransomts.github.io/ASEE-2026/

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`https://ransomts.github.io/ASEE-2026/`

Encourage the audience to scan the QR code and explore the data themselves. The interactive tool shows everything—not just the results we selected for the paper.

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Key Takeaways

- STEM domains are interdisciplinary in *structurally different ways*
  - Engineering: parallel (shared core, divergent specialization)
  - Computing: nested hierarchies from CS
  - Sciences: cross-boundary clusters
- One-size-fits-all policy won't work

Key Takeaways

- STEM domains are interdisciplinary in *structurally different ways*
  - Engineering: parallel (shared core, divergent specialization)
  - Computing: nested hierarchies from CS
  - Sciences: cross-boundary clusters
- One-size-fits-all policy won't work

Synthesis slide. The structural difference between domains is the main takeaway. The Aerospace–Industrial convergence is invisible to departmental taxonomy—multi-metric analysis is needed.

Practical Implications

- Nested (IT $\subset$ CS): guide students toward parent field's electives
- Parallel (engineering): coordinate shared prerequisites
- Convergent (Aero $\approx$ Ind Eng): cross-listed courses, certificates, dual degrees

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Give the audience something actionable. These aren't just academic categories—they suggest specific interventions at the advising, departmental, and university level.

Future Work

- Disaggregate required vs. elective coursework
- Time series: are programs converging or diverging?
- Institutional disaggregation across universities
- Validate against known interdisciplinary programs

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- Time series: are programs converging or diverging?
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Acknowledge limitations honestly. The biggest one is that we can't currently separate required from elective enrollment.

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Thank the audience. Remind them of the QR code for the interactive explorer.